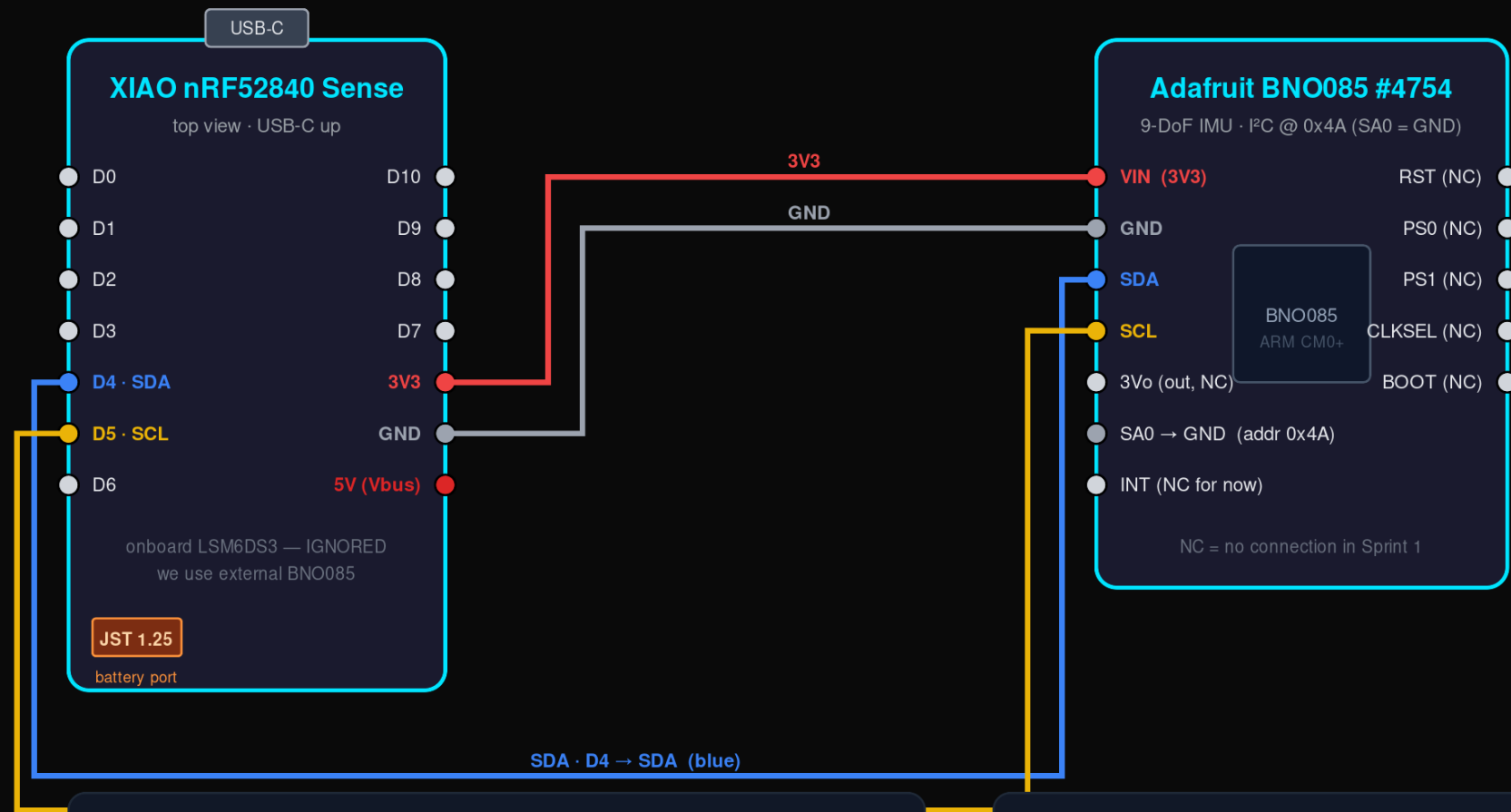


AI Sport Monitor — Wiring Diagram

XIAO nRF52840 Sense · Adafruit BNO085 (#4754) · LiPo 3.7 V 400 mAh
Sprint 1 — running node · I²C @ 400 kHz · 100 Hz sampling

Cable color code

3V3	GND	SDA	SCL
BAT +	BAT -		



Battery — LiPo 3.7 V 400 mAh · format 602030



Polarity check before plugging — Adafruit / Seeed cells use opposite pinout on some batches.

Build notes

- I²C address: 0x4A — SA0 pin tied to GND on the BNO085 breakout.
 - I²C speed: Wire.setClock(400000) — 400 kHz fast mode required for 100 Hz.
 - Power: 3V3 from XIAO (LDO) → BNO085 VIN. Do NOT use 5V (Vbus).
 - BNO085 INT not connected — the driver polls via I²C in Sprint 1.
 - Battery: LiPo connects to XIAO's onboard JST. Charging happens through USB-C.
 - Onboard LSM6DS3 in the XIAO Sense is intentionally ignored — we need 9-DoF.
- Doc rev. Sprint 1 · matches firmware/running_node/config.h UUIDs and pinout.